



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Let's talk about data storage!



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Ask your partner:

- What ways of storing data do you know?
- Is it better to have historical data or is it better to have transactional data of a company? Why?

Can Do Statements:

- I can compare data warehouses and operational databases.
- I can give examples related to data warehouses and operational databases.
- I can describe key features related to data warehouses and operational databases.



Let's Explore!

The teacher compares data warehouses and operational databases.

DATABASE



Instructions: Read the text below.

Student: Hello, teacher! Can you tell us whether we are going to continue with the next topics or not?

Teacher: You cannot continue with the contents of the program if you don't know about data warehouses and operational databases, so today we will talk about them.

Student: Okay, proceed.

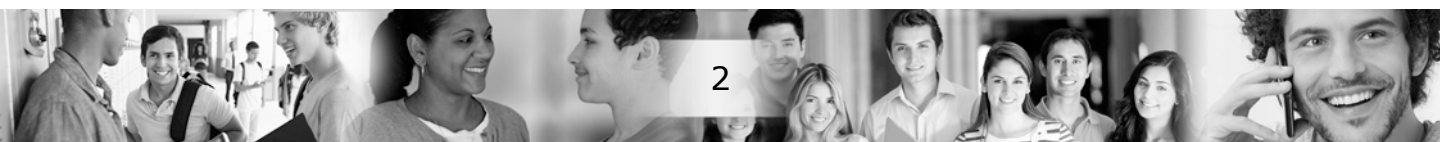
Teacher: Well, for starters, a data warehouse is where an organization or company keeps information electrically. Some of the key features of a data warehouse are that you can use it for reporting and for the analytical processing of data; so, it organizes data in a subject-oriented manner. Revenue can be an example of a data warehouse, for it incorporates data from other operational systems that contain information like student tuition data, alumni gift data, financial aid data.

Student: And operational databases?

Teacher: Alright, you don't have to use a data warehouse if you don't want to. That's why you have operational databases. An operational database is the one that allows users to access it and update it on a regular basis, its key features are that it has to deal with the transaction processing of the business. The approach that operational databases use is OLTP which means On Line Transaction Processing Databases. Let me give you an example. If a company wants to track stock and see how many items of something they have available, they use an operational database because they can see all that information in real time.

Student: You should tell us whether there are more similarities or differences between these two concepts.

Teacher: Absolutely! Let me give you a brief comparison of the two. They are similar, for both of them are databases and you can use them for the same purpose: storage and organization of data. Now, some differences are that data warehouses have historical data which facilitates analysis, but operational databases only have current data, so they only concentrate on the data being recorded as of now. Another difference is that you don't have the opportunity to change or delete data if you use data warehouses, but with operational databases, you can modify or even delete data.



**Let's Practice!**

Let's compare databases!

Instructions: Complete the sentences by writing its corresponding word. Use capital letter where necessary. Use: so they, so, for, or.

1. They are similar, ____ both of them are databases and have the same purpose: storage and organization of data.
2. Data warehouses have historical data, ____ it facilitates analysis but it doesn't allow changing or deleting information.
3. Operational databases only have current data, ____ only concentrate on the data being recorded as of now.
4. Whether you use data warehouses ____ operational databases you will always be able to analyze the data in them.

**Let's Practice!**

Where do you see data warehouses and operational databases?

Instructions: Proofread the sentences. Correct the mistakes.

1. Revenue, can be an example of a data warehouse from eat corporates data from other operational systems that contain information.
2. An example of operational databases are the daily transactions that we do in business, whether you use the usual approach called OLTP.
3. Data warehouses contain a read-optimized approach, or it contains information like student tuition data, alumni gift data, and financial aid data.





Extension Activity

Let's learn how to use data warehouse and operational database

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search more information and examples about data warehouse and operational database.
- Identify the main steps to create a database using the ones from your research.
- Create a data warehouse and operational database with any information.
- Prepare a presentation in which you explain the process of creating a data warehouse and an operational database.
- Check your pronunciation with your instructor.

Self-Assessment

- | | Yes | No |
|--|-----------------------|-----------------------|
| 1. I look for information and examples about data warehouse and operational database. | ☐ | ☐ |
| 2. I identify the main steps to create a database. | ☐ | ☐ |
| 3. I create a data warehouse and operational data base. | ☐ | ☐ |
| 4. I prepare a presentation explaining the process of creating a data warehouse and an operational database. | ☐ | ☐ |
| 5. I check my pronunciation with my instructor. | ☐ | ☐ |



Target Words

- **proceed** (verb): to continue one's discourse
e.g. Teacher: You cannot continue with the contents of the program if you don't know about data warehouses and operational databases, so today we will talk about them.
Student: Okay, proceed.
- **for starters** (idiom): to start with, initially
e.g. Well, for starters, a data warehouse is where an organization or company keeps information electrically.
- **revenue** (noun): the amount of money that a company receives during a specific period of time
e.g. Revenue, can be an example of a data warehouse, for it incorporates data from other operational systems that contain information like student tuition data, alumni gift data, financial aid data.

- **absolutely** (interjection): it is used when you want to express complete agreement
e.g. Absolutely! Let me give you a brief comparison of the two.
- **tuition** (noun): the charges you have to pay when you study in a private school or university
e.g. Revenue, can be an example of a data warehouse, for it incorporates data from other operational systems that contain information like student tuition data, alumni gift data, financial aid data.
- **all right** (adverb): similar to yes, very well, OK
e.g. Alright, you don't have to use a data warehouse, if you don't want to. That's why you have operational databases.



Learning Points

Conjunctions

In the English language, there are some words that are called “conjunctions”. A conjunction is a word that you can use to connect parts of speech like: other words, phrases or sentences. **For**, **so**, **whether** and **or** are common conjunctions.

For is a preposition, but you can also use it as a conjunction. It has a similar meaning as that of **because** or **since**.

Example:

*John thinks he is a good programmer, **for** he studied several years to be one.*

So is another word that you can use as a conjunction as well, it has a similar meaning to **therefore** or **consequently**.

Example:

*Mary won the lottery, **so** she will be buying lots of things.*

Whether and **or** are two conjunctions that people commonly use together or in the same sentence, and they express a doubt or choice between two possibilities.

Example:

*I want you to tell me **whether** to use data warehouse **or** operational databases.*

Examples

- *Can you tell us whether we are going to continue with the next topics or not?*
- *Revenue, can be an example of a data warehouse, for it incorporates data from other operational systems that contain information like student tuition data, alumni gift data, financial aid data.*
- *Some of the key features of a data warehouse are that you can use it for reporting and for the analytical processing of data; so, it organizes data in a subject-oriented manner.*





Learning Points

Zero Conditional Starting With the Main Cause in Negative

The zero conditional is one of the four types of conditionals we have in English. You use it to talk about a real situation (condition) and the result of that situation (result). This conditional has two parts; the **if clause** and the **main clause** or **result**.

We can use the zero conditional starting with the main clause or the result in negative. Remember we need to use the present simple here, so the structure would go like this:

Main clause (negative), present simple	"If" clause, present simple
<i>You don't use databases correctly</i>	<i>If you don't read enough information about them.</i>

Note:

When you start a conditional with the main clause, you don't need to use a comma to separate the clauses.

Example:

*You **don't** have to continue using this database **if** you do not read more about it.*

Examples

- You don't have to use a data warehouse if you don't want to. That's why you have operational databases.*
- You don't have the opportunity to change or delete data if you use data warehouses.*

